

Hierarchical Inheritance

Basic Information

Title: Hierarchical Inheritance

File Name: HierarchicalInheritanceDemo.java

Objective / Aim

To demonstrate hierarchical inheritance in Java, where multiple child classes inherit from one common parent.

System Requirements

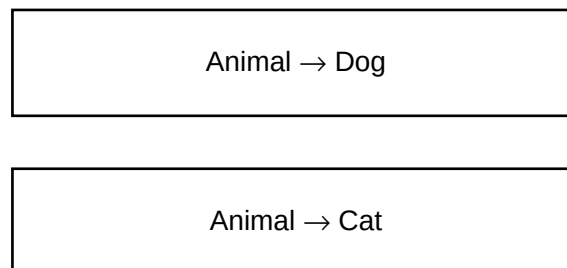
Hardware: Computer/Laptop, minimum 4 GB RAM, keyboard and mouse.

Software: Java Development Kit (JDK), Java compiler, Java Runtime Environment (JRE), and any Java-supported IDE or terminal.

Algorithm / Logic Description

1. Start the program.
2. Create Animal as the common parent class.
3. Create Dog extending Animal.
4. Create Cat extending Animal.
5. Create objects of Dog and Cat.
6. Call the common and child-specific methods.
7. End the program.

UML Class Diagram



Source Code

```
// Common parent class
class Animal {

    void eat() {
        System.out.println("Animal is eating");
    }
}

// First child class
class Dog extends Animal {

    void bark() {
        System.out.println("Dog is barking");
    }
}

// Second child class
class Cat extends Animal {

    void meow() {
        System.out.println("Cat is meowing");
    }
}

public class HierarchicalInheritanceDemo {
    public static void main(String[] args) {

        Dog dog = new Dog();
        Cat cat = new Cat();

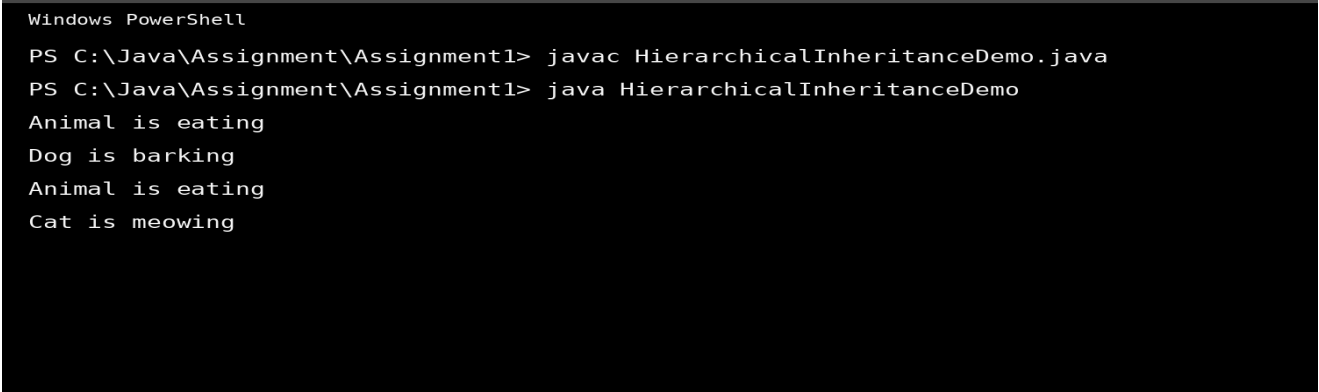
        // Both children inherit eat()
        dog.eat();
        dog.bark();

        cat.eat();
        cat.meow();
    }
}
```

Compilation & Execution Commands

```
javac HierarchicalInheritanceDemo.java
java HierarchicalInheritanceDemo
```

Output Screen



```
Windows PowerShell
PS C:\Java\Assignment\Assignment1> javac HierarchicalInheritanceDemo.java
PS C:\Java\Assignment\Assignment1> java HierarchicalInheritanceDemo
Animal is eating
Dog is barking
Animal is eating
Cat is meowing
```

Result

Thus, the Hierarchical Inheritance program was implemented successfully using Java inheritance concepts.